HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

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Formal verification
Verification of a arithmetic logic unit (ALU)
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Module to verify

We are interested in the verification of an arithmetic logic unit able to manage 8 operations, on two input vectors.
A generic parameter `SIZE` allows to define the operands size.
The input/output are :

Nom	Taille	Description
`a_i`	SIZE	First operand
`b_i`	SIZE	Second operand
`mode_i`	3	Operation mode
`c_o`	1	Carry out
`s_o`	SIZE	Result

The 3-bit binary value of `mode_i` selects the operation shown below. The expected results are :

mode_i	Operation	**s_o**	**c_o**
`"000"`	ADD	`a_i+b_i`	carry out
`"001"`	SUB	`a_i-b_i`	borrow
`"010"`	OR	`a_i or b_i`	not specified
`"011"`	AND	`a_i and b_i`	not specified
`"100"`	PASS A	`a_i`	not specified
`"101"`	PASS B	`b_i`	not specified
`"110"`	EQUAL	`s_o(0)= '1' when a_i=b_i else '0'`	not specified
`"111"`	ZERO	`0`	not specified

The mathematical operations are computed with unsigned numbers.
For subtraction, `c_o` is high when the unsigned subtraction borrows. For operations other than ADD and SUB, `c_o` is not specified. For EQUAL, only `s_o(0)` is specified; the upper result bits are not specified.
A second generic parameter `ERRNO` is used to test the assertions. Its behavior is the following :

1. When in the $[0, 15]$ interval the result is valid;
2. When in the $[16, 24]$ interval, the result is invalid.

The provided SBY script tests `ERRNO` values 0, 1, 3, 7, 9, and 12 as valid configurations, and values from 16 to 24 as errors for which it is expected to fail.
You can modify its value in the `alu.sby` file.